
GENERAL RELATIVITY

Tutorial problem set 8, 01.11.2019.

■ PROBLEM 22 Killing vectors.

- (a) Show that the commutator of two Killing vectors is a Killing vector. Show that a linear combination with constant coefficients of two Killing vectors is a Killing vector.
- (b) Show that Killing vectors satisfy the following identities

$$(i) \quad \nabla_\mu \nabla_\nu K^\rho = R^\rho{}_{\nu\mu\sigma} K^\sigma, \quad (22.1)$$

$$(ii) \quad \nabla_\mu \nabla_\nu K^\mu = R_{\nu\mu} K^\mu, \quad (22.2)$$

$$(iii) \quad K^\alpha \nabla_\alpha R = 0. \quad (22.3)$$

Hints: For (i) first show that for any vector V^ρ

$$[\nabla_\mu, \nabla_\sigma] V_\rho = R_{\rho\nu\mu\sigma} V^\nu, \quad (22.4)$$

and make use of Killing's equation and a properties of the Riemann tensor. For (iii) make use of one of Bianchi identities from last time and the fact that the 2-tensor can always be decomposed into a symmetric and an antisymmetric part.

■ PROBLEM 23 Killing vectors on a 2-sphere.

The two-sphere is given by

$$ds^2 = d\vartheta^2 + \sin^2 \vartheta d\varphi^2. \quad (23.1)$$

- (a) Solve the Killing's equation to find that the Killing vectors of a 2-sphere are

$$K_1 = \cos \varphi \partial_\vartheta - \cot \vartheta \sin \varphi \partial_\varphi, \quad (23.2)$$

$$K_2 = -\sin \varphi \partial_\vartheta - \cot \vartheta \cos \varphi \partial_\varphi, \quad (23.3)$$

$$K_3 = \partial_\varphi. \quad (23.4)$$

- (b) Show that the commutators of these three Killing vectors satisfy the following commutation relations

$$[K_1, K_2] = K_3, \quad (23.5)$$

$$[K_2, K_3] = K_1, \quad (23.6)$$

$$[K_3, K_1] = K_2. \quad (23.7)$$

Check that they satisfy the algebra of Lie groups. Can you see which algebra is it?

■ **PROBLEM 24** Maximally symmetric spaces.

The metric of a *de Sitter space* is given by

$$ds^2 = \frac{1}{\cos^2(H\eta)} \left[-d\eta^2 + \frac{d\chi^2}{H^2} + \frac{\sin^2(\chi)}{H^2} (d\vartheta^2 + \sin^2\vartheta d\varphi^2) \right] \quad (24.1)$$

- (a) Calculate the Riemann tensor, the Ricci tensor and the Ricci scalar.

Note: For a diagonal metric Christoffel symbols can be calculated using the following simplified formulas

$$\begin{aligned} \Gamma_{\alpha\beta}^\gamma &= 0, & \Gamma_{\alpha\alpha}^\beta &= -\frac{1}{2g_{\beta\beta}} \partial_\beta g_{\alpha\alpha}, \\ \Gamma_{\alpha\beta}^\beta &= \partial_\alpha \left(\ln \sqrt{|g_{\beta\beta}|} \right), & \Gamma_{\alpha\alpha}^\alpha &= \partial_\alpha \left(\ln \sqrt{|g_{\alpha\alpha}|} \right), \end{aligned} \quad (24.2)$$

where $\alpha \neq \beta \neq \gamma \neq \alpha$ and there is *no* summation over indices.

- (b) Show that

$$R_{\rho\sigma\mu\nu} = \frac{R}{D(D-1)} (g_{\rho\mu}g_{\sigma\nu} - g_{\rho\nu}g_{\sigma\mu}), \quad (D=4). \quad (24.3)$$

A space for which this last equation is satisfied is called a maximally symmetric space. In such a space, the curvature is the same everywhere and the same in every direction. Hence, if one knows the curvature in one point of the space, it is known everywhere.